Khushi Teli

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Summary

Data engineer and M.S. Data Science candidate at Boston University building **Python/SQL** ETL and AI data services. Landed 1M health events in **PostgreSQL** with data-quality checks; built a **RAG** assistant over census data.

Education

Boston University — Boston, MA
Master of Science, Data Science — *Expected Jan 2027*

Vellore Institute of Technology — Vellore, TN, India
Bachelor of Technology, Computer Science and Engineering — *2021 – 2025*

Experience

Graduate Research Assistant — Sep 2026 – Present
Boston University — *Boston, MA*
- Benchmarked Red Hat's **asago** risk extraction on security policies against expert keys using **precision/recall/F1**
- Deploying **vLLM** inference on **OpenShift** (Mass Open Cloud) to test asago on real university security policies
- Diagnosed failure modes (dependency conflicts, token truncation, risk over-inclusion) and gave maintainers fixes

Software Engineering Collaborative Partner — May 2026 – Aug 2026
Red Hat — *Boston, MA*
- Architected 11-stage **Kubeflow** pipeline on **OpenShift AI** producing documentaries; cut production time 85%
- Productionized modular components with automated error handling for unattended runs on **OpenShift AI**
- Chained speech recognition, **LLMs** and TTS to automate narration, saving 10+ hours per interview

Teaching Assistant, Foundations of Data Science using Python — Mar 2026 – May 2026
Boston University — *Boston, MA*
- Held weekly office hours helping students debug **Python**, **pandas** and **NumPy** code for course assignments
- Graded assignments and gave written feedback on data cleaning, visualization and exploratory analysis

Data Engineer (Client: GBH) — Mar 2026
Boston University — *Boston, MA*
- Built **Python** ETL pipelines unifying 10+ **ACS** datasets into 20+ socioeconomic indicators for 260 MA towns
- Developed **Gemini RAG** assistant over 12 years of census data; cut report retrieval from hours to seconds

Data Analyst (Client: MuckRock) — Sep 2025 – Dec 2025
Boston University — *Boston, MA*
- Cleaned, validated and merged 100K+ public records from multiple sources in **Python** for trend analysis
- Built **Datasette** dashboards and **SQL** reporting on U.S. military-base slot-machine revenue, FY2020–2024

Projects

Health Events ETL Pipeline | *Python, SQLAlchemy, PostgreSQL, MySQL, Flask, OpenShift*
- Built idempotent **Python/SQLAlchemy** ETL landing 1M health events in **PostgreSQL** with batched upserts
- Added null, type, enum and dedup data-quality checks with JSON logging and a deterministic **SQL** priority score
- Migrated curated data to a **MySQL** serving store with parity checks; served it via **Flask** on **OpenShift**

Drug-Associated Acute Kidney Injury Risk Prediction | *BigQuery SQL, Python, CatBoost, SHAP* | **Link**
- Built 65K+ ICU-stay cohort from **MIMIC-IV** in **BigQuery SQL**, linking nephrotoxic drugs to 48-hr AKI
- Trained **CatBoost** with **SHAP**; nearly doubled recall (0.37 → 0.77) and lifted ROC-AUC 0.82 → 0.85

Hackathon Wins

Challenge 2 Winner, MedAI Hackathon | *Boston University, Hariri Institute* — Feb 2026
- Predicted acute tubular injury risk before biopsy from 6,500+ protein biomarkers and clinical data

Original Research Award, CivicHacks | *Mapping for Good: The Boston Food Ecosystem* — Apr 2026
- Built a neighborhood-level food access platform joining Boston Open Data, **ACS** and social vulnerability data

Technical Skills

Languages & Data: Python, SQL, PostgreSQL, MySQL, SQLAlchemy, BigQuery, Spark, MongoDB, R
Cloud & Pipelines: AWS, GCP, Git, Docker, Kubernetes, Kubeflow, OpenShift AI, Flask
LLMs & NLP: RAG, vLLM, Gemini API, Transformers
ML & Analytics: scikit-learn, PyTorch, TensorFlow, CatBoost, SHAP, Statistical Modeling, Tableau, Power BI